2 Group 17 elements form a range of oxoacids with different oxidation states, such as $HClO_4$.

(a) **(i)** $HClO_4$ has two central atoms; one chlorine atom and one oxygen atom. In addition, the H atom is bonded to a O atom. Draw the dot-and-cross diagram of $HClO_4$.

[1]

(ii) Use VSEPR to describe and explain the shape and bond angle about central *Cl* and O atom in $HClO_4$.

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..... [3]

(b) With reference to your answer in **(a)(i)**, explain why HFO_4 does not exist?

.....

..... [1]

(c) The Latimer diagram of some chlorine species in acidic solution is given in Fig. 2.1.

$$\underset{+1}{HOCl} \xrightarrow{+1.64\ V} \underset{0}{Cl_2} \xrightarrow{+1.36\ V} \underset{-1}{Cl^-}$$

Fig 2.1

In Latimer diagram, oxidation numbers decrease from left to right and the numerical values of $E^\ominus$ of two adjacent species in volts. For example, the diagram shows that $E^\ominus(HOCl|Cl_2)$ is +1.64 V.

(i) Write a half-equation for the conversion of one mole of $HOCl$ to one mole of Cl^- in acidic solution at 298 K.

..... [1]

(ii) Hess' Law is applicable to $\Delta G^\ominus$ in the same manner as $\Delta H^\ominus$.
With reference to the *Data Booklet* and Fig. 2.1, calculate $\Delta G^\ominus$ for the conversion of one mole $HOCl$ to one mole of Cl^- in acidic solution.

[3]

(iii) Hence, calculate the standard electrode potential for the $HOCl|Cl^-$ half-cell.

[1]

(d) A sample of group 2 metal ethanedioate, MC_2O_4, is analysed to determine the identity of the metal.

A 4.13 g sample of the metal ethanedioate is reacted with excess acidified potassium manganate(VII), $KMnO_4$.

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

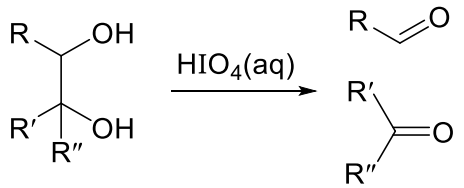
A 25 cm^3 portion from the remaining solution is reacted with periodate ions, IO_4^-, to produce IO_3^- and MnO_4^-. 22.2 cm^3 of 0.2 mol dm^{-3} of IO_4^- was required to react with the manganese(II) ions.

$$2Mn^{2+} + 5IO_4^- + 3H_2O \rightarrow 2MnO_4^- + 5IO_3^- + 6H^+$$

Deduce the identity of the metal in the metal ethanedioate.

[3]

(e) Periodic acid, HIO_4, is used as a selective oxidant in organic chemistry to split alcohols with two adjacent hydroxy groups into two carbonyl compounds.
An example is shown in Fig. 2.2.



R, R′, R″ = H or alkyl

Fig. 2.2

Periodic acid also oxidises hydroxycarbonyl and dicarbonyl compounds by a hydration equilibrium, in which the carbonyl group is first converted into a diol.

$$R_2C{=}O + H_2O \rightleftharpoons R_2C(OH)_2 \qquad R = H \text{ or alkyl}$$

For example:

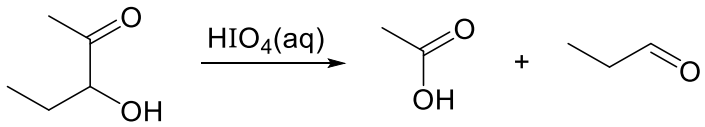


Fig. 2.3

Predict the organic products of the reactions shown in Fig. 2.4.

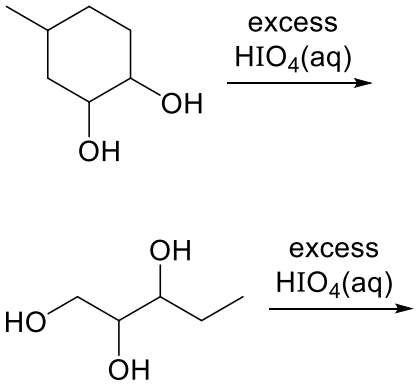


Fig. 2.4

[3]

[Total: 16]